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SUPERSEDING
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6 June 1968

MILITARY SPECIFICATION

SONOBUOYS, DETECTION DEVICES, PRESERVATION, PACKAGING AND PACKING, PROCEDURES FOR

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 **This specification covers the requirements for preservation,** packaging and packing of air launched sonobuoys and detection devices. (See 6.1.)

2. APPLICABLE DOCUMENTS

2.1 **The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.**

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	**Preservation, Methods of**
MIL-D-3464	**Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification**
MIL-C-5541	**Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys**
MIL-T-43036	**Tape, Pressure-Sensitive Adhesive, Plastic Film, Filament Reinforced (For Sealing Fiber Containers and Cans)**

FSC PACK

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STANDARDS

Federal

Fed. Test Method Std. No. 101	Preservation, Packaging, and Packing Materials, Test Procedures

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MS20003	Indicator, Humidity, Card, Three Spot, Impregnated Areas (Cobaltous Chloride)
MS33586	Metals, Definition of Dissimilar

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government quality control representative. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Forms and Publications Center (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article inspection - This specification makes provision for first article inspection of the container conforming to this specification (see 4.3). Drawings, when specified on applicable DD Form 1423 (see 6.2.1), showing design, including all dimensions and materials shall be furnished and shall comprise a part of the first article inspection requirements. The first article samples furnished shall be tested and pass the first article tests specified herein.

3.2 Materials -

3.2.1 Metals - Bare metals shall be of the corrosion-resistant type or suitably treated to resist deterioration due to aviation gasoline and jet fuels, salt spray, and atmospheric conditions likely to be met in storage or normal service. Non-corrosion resistant steel fittings, if uncoated, shall be cadmium plated in accordance with QQ-P-416, Class 3. Aluminum parts shall be chemically treated in accordance with MIL-C-5541, Class 3.

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3.2.2 Dissimilar metals - Dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.2.3 Non-metallic materials - Plastic, glass fiber reinforced plastics, or other compounded non-metallic materials may be used for the container provided they are suitable for the purpose and can meet all the performance requirements. If in contact with metallic surfaces, these non-metallic materials shall not induce or promote corrosion of the metallic components.

3.2.4 Protective coatings - If protective coatings are used, they shall not block, crack, flake, craze or alligator.

3.2.5 Desiccant - Desiccant in the quantity determined by MIL-P-116, Method II, Formula II is required and shall conform to MIL-D-3464, Type II. The desiccant shall be distributed in any area of the container where it will not impair its function and yet provide the required desiccation.

3.2.6 Humidity indicator card - A humidity indicator card conforming to MS20003 shall be placed so as to be readily visible upon opening, but it shall not contact either the desiccant or metal surface of the sonobuoy or detection device.

3.2.7 Cushioning material - Cushioning material, when used, shall provide the required shock protection but shall be non-corrosive to the sonobuoy or detection device. Expanded or foamed plastics or other suitable cushioning materials meeting the requirements above shall be used.

3.2.8 Collars - Collars (or inserts) if used to assist the immobilization of the sonobuoy or detection device shall be fabricated of non-corrosive material.

3.2.9 Aging - Materials fabricated into the container shall not be adversely affected by aging deterioration to an extent which significantly impairs performance during the intended service life of 5 to 7 years as required in the item specifications.

3.3 Construction - Containers shall be such as to provide protection for the sonobuoy or detection device against shock and deterioration during storage and shipment. The containers shall be of any configuration that will keep cube to a minimum. The outside diameter or lateral dimension across flats, if not cylindrical, shall not exceed 7 inches. The specified sonobuoy or detection device shall be a snug fit in the container, with cushioning as necessary, but shall not require undue effort for insertion or removal of the item. The container and cushioning shall be of minimum practicable weight consistent with the performance requirements of this specification. The material thickness of the container as shown on the manufacturer's drawing shall show a tolerance which shall be not greater than ±10 percent.

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3.3.1 Closure - An air tight closure, readily reclosable and resealable with simple fasteners and capable of being opened without special tools shall be provided. If pressure-sensitive tape is used, it shall be accomplished with pressure-sensitive tape conforming to MIL-T-43036 or any means which will allow the closure to pass the required tests.

3.4 Physical properties - When tested as specified in Section 4, the containers shall conform to the requirements specified in Table I.

3.4.1 Functional tests - After the Series II tests the item shall pass all functional tests as noted in the sonobuoy or detection device specification.

3.4.2 Accelerated environmental test - There shall be no external evidence of deterioration or effect which indicates that the packaging system will not provide adequate protection for the sonobuoy or detection device during service. Evident failure during any phase shall be justification for discontinuance of further testing. During the "leaks in container" phase the container shall not show a steady stream or recurring succession of bubbles from any surface or seam. After Series II tests, the sonobuoy or detection device and the interior packaging such as indexing, cushioning and other shock attenuating media, and interior barriers if used, and the container, shall show no visible injury that would affect their utility. There shall be no evidence of shifting within the container which would create conditions likely to cause such damage during shipment. Waterproof papers, barriers, wrappings, and containers shall be intact and capable of providing the intended protection. There shall be no separation or dislocation of container or component container parts. There shall be no evidence of dents on the skin of the sonobuoy or detection device. There shall be no evidence of cracking or separation of any external coatings. The sonobuoy or detection device shall be removable, without undue effort, from the container.

3.5 Unitizing procedures for containers -

3.5.1 Grouping - Sonobuoy or detection devices, each in unit containers, shall be grouped in quantities of 2 or 4 depending upon weight as noted below. Sonobuoys or detection devices weighing 25 pounds or less shall be grouped four each and those weighing over 25 pounds shall be grouped as two each. The unit containers shall be secured into these groups, so as to prevent shifting, by means of weather-resistant, high strength tape or other non-shifting banding material suitable for the purpose.

3.5.2 Palletizing - Groups of two or four unit containers, as applicable, shall be palletized and secured in an upright condition on 4-way entry, double wing pallets, so that there are 48 unit containers on a pallet. The pallet size shall not exceed 43-1/2 inches by 57-1/2 inches, and shall incorporate a wood cap over the unitized groups to secure and protect the palletized load. Metal strapping shall be utilized to secure the caps tightly to the pallet. A suggested construction is noted in

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TABLE I

PHYSICAL PROPERTIES

TEST		SPECIMEN	REQUIREMENTS	TEST PARAGRAPH
SERIES I (in sequence as below)				
Drop test (free fall)		Grouped packaged sonobuoys or detection devices (3.5.1), 2 or 4 to group, as applicable, and banded or strapped, as for shipment	No evidence of substantial racking, loosening of strapping, or shifting of individual containers	4.5.1.1
Water vapor transmission rate		All individual containers from group above	0.05 gms/100 sq. in./ 24 hr. (max.)	4.5.1.2
SERIES II (Complete sequence below)		New group, all individual containers in this group to be tested except as noted below	See below and paragraph 3.4.2	4.5.2
Accelerated environmental test				
Repeat four times	3 hours at 160° F 3 hours at -65° F	All containers from group above		4.5.2.1
	1 hour simulated rainfall	All containers from group above	—	4.5.2.1.1
	16 hours ultraviolet exposure	All containers from group above	—	4.5.2.1.2
	2 hours at 160° F 2 hours at -25° F	All containers from group above	—	4.5.2.1.3
	Tipover test	All containers from group above	—	4.5.2.2
	Drop test (free fall)	All containers from group above	—	4.5.2.3
	Group tipover test	All 2 or 4 containers banded together as a group	—	4.5.2.4
	Simulated mechanical impact test (alternate test to Group tipover test)	All containers from group above		4.5.2.4.1
	Leaks in container test. (Hot water technique)	All containers from group above	No continuous evolution of bubbles from any surface or presence of water inside after opening	4.5.2.5

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Section 6 (see 6.4). There shall be no more than three adjacent frequency channels unitized and shipped on a pallet and the packing list shall be attached only to item one of a shipment (see 3.7.2). All additional odd containers (in excess of multiples of two or four of a particular frequency channel) shall be shipped intermingled on one pallet.

3.6 Stacking - Palletized containers shall be suitable for stacking one upon another, in such a manner as not to interfere with forklift access. Each palletized load shall be capable of supporting two similar pallet loads (with contents) in a superimposed position without damage to the containers.

3.7 Marking - In addition to any special marking required by contract or order, containers shall be marked in accordance with the requirements of MIL-STD-129. This shall include fragile marking, applicable lot number, and Level A marking.

3.7.1 Special handling instructions - In addition to the above markings, to each group of two or four unit containers a transparent plastic packing envelope shall be securely attached containing a card with the following instructions:

SPECIAL PRECAUTIONS

FRAGILE. HANDLE WITH SAME CARE AS GIVEN ANY SENSITIVE ELECTRONIC EQUIPMENT.

MAINTAIN FIRM GRIP WHILE HANDLING.

DO NOT DROP OR DAMAGE THESE CONTAINERS OR ALUMINUM BUOY HOUSING IN ANY WAY.

BE SURE SCUTTLING PLUG IS NOT DAMAGED IN ANY WAY.

3.7.2 Packing list - The packing list shall be attached to the two or four unit pack containing the lowest serial number of a shipment.

3.7.3 Method II label - Precautionary label for Method II of MIL-STD-129 shall be applied.

3.7.4 Frequency channel marking - The type and frequency channel number shall be marked on top and bottom surfaces of each container. In addition, these markings shall also appear lengthwise on three locations on the longitudinal surface, approximately 120 degrees apart.

3.7.5 Pallet numbering - The pallets shall be numbered, progressing in order from frequency channel one through the highest frequency channel shipped.

3.8 Workmanship - The containers shall be free from any defects which may affect their durability, strength and serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The inspection and testing of containers shall be classified as follows:

(a) First article inspection - First article inspection consists of those examinations and tests conducted on a container representative of those to be furnished under the contract, for the purpose of determining whether all requirements of applicable specifications are being met (see 4.3).

(b) Quality conformance inspections - Quality conformance inspection consists of those visual examinations and tests performed during the course of production on individual lots which have been submitted for Navy acceptance (see 4.4).

4.3 First article inspection - First article inspection shall consist of all the examinations and tests of this specification. Examinations and tests shall be conducted at the contractor's plant or any other location acceptable to the Government (see 6.3).

4.3.1 First article inspection samples - The first article inspection samples shall consist of complete unit containers that are representative of production items. The contractor shall conduct tests on the complete containers to ascertain that the packaging system conforms to this specification.

4.3.1.1 Two groups of four or four groups of two each of the containers, as applicable, assembled as for shipment, shall be tested. One group of four each shall be subjected to (Series I), first as a strapped or banded group for the free fall drop test, and then each container from the group shall be tested for water vapor transmission rate. The individual containers from the second group shall be subjected to all phases of the Accelerated environmental test (Series II). When four groups of two each of the containers assembled as for shipment, are submitted, two groups shall be subjected to the free fall drop test and then each of the individual containers shall be tested for water vapor transmission rate. The individual containers (4) from two other groups shall be subjected to the Accelerated environmental test (Series II).

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